| Problem Chosen | 2025 HiMCM Summary Sheet | Team Control Number |
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## Summary

In recent years, while major sporting events have generated substantial economic and social benefits, their significant environmental footprints have attracted increasing attention. In response, this study evaluates the environmental impacts of large-scale sporting events and develops a quantitative model capable of ranking host cities by sustainability performance.

To quantify the environmental footprint, we adopt a hierarchical decomposition approach: the overall impact is first divided into four dimensionsenergy consumption, greenhouse gas emissions, water use, and waste generationand. Each dimension is further broken down into its key drivers, including venue operations, broadcasting, transportation, lodging, food and beverage services, and cleaning and maintenance. Then, we construct detailed sub-models for energy (three sub-factors), greenhouse gas emissions (four sub-factors), water use (four sub-factors), and waste generation (four sub-factors). To ensure consistent and manageable environmental accounting, the model applies a host-city-restricted system boundary. Applying the model to Super Bowl LIX yields an estimated energy use of 426.7 MWh, water consumption of 15,323 t, greenhouse gas emissions of 825.8 t $CO_2$e, and 75.2 t of waste, validating the accuracy and applicability of the proposed framework.

Building on the environmental quantification, we develop a multi-criteria sustainability evaluation method that integrates the entropy weight method with TOPSIS. The entropy method assigns objective weights to each environmental indicator, while TOPSIS ranks cities based on their distance to the positive and negative ideal solutions. Results show that among six previous Super Bowl host cities (Glendale, Las Vegas, New Orleans, Santa Clara, Inglewood, and Atlanta), New Orleans achieves the highest sustainability score. Among four NFL home cities that have not yet hosted the Super Bowl (Houston, Minneapolis, Indianapolis, and Charlotte), Indianapolis ranks first, demonstrating strong potential as a future host.

Extending beyond the single-city, single-game structure of the Super Bowl, we further adapt the model to multi-stage, multi-city single-sport tournaments. By comparing structural differences between single-event and multi-stage formatsparticularly in spatial dispersion, team mobility, and resource-load distributionwe incorporate stage-wise resource accumulation, inter-city travel emissions, event frequency, venue reuse, and field-maintenance water use into the extended framework. Using the BWF World Tour as an illustrative case, we compare route configurations from the 2018, 2019, 2020, and 2021 seasons and find that the 2018 route demonstrates the best overall environmental performance.



**Keywords**: Sport mega-events; Environmental footprint assessment; Entropy weight method; TOPSIS method; Comprehensive evaluation model

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# 1 Introduction

Sport mega-events such as the Super Bowl have become central spectacles in modern society, drawing the attention of millions worldwide and generating enormous social and economic value. These events serve as cultural celebrations that unite fans and stimulate local economies through tourism, sponsorship, and media exposure. However, beneath their glamorous appearance lies a substantial environmental cost [1]. From massive air travel and hotel accommodations to stadium energy consumption, food waste, and broadcasting operations, sport mega-events create significant ecological footprints that cannot be ignored.

As global awareness of climate change and sustainability rises, the environmental performance of these large-scale sporting events has attracted growing scrutiny. Hosting cities face increasing pressure to balance economic benefits with environmental responsibilities. The Super Bowl, as one of the most influential single-sport events, consumes vast resources within a short period, creating short-term spikes in energy usage and greenhouse gas emissions. On the other hand, tournaments such as the FIFA World Cup or the Olympic Games exert sustained environmental impacts across multiple venues and longer durations. Therefore, it is imperative to develop comprehensive models to evaluate and reduce the environmental footprint of such events, ensuring that future sport mega-events can be conducted in a more sustainable and responsible manner.

In this paper, we explore the environmental impact of sport mega-events and try to find better, more sustainable host choices. Our work is organized as follows.

| Task | Problem Restatement | Organization |
|---|---|---|
| 1 | Analyze key environmental impact factors of mega sporting events and establish an evaluation model. | Section 2 |
| 2 | Develop a sustainability-based decision model and apply it to determine optimal host cities under different scenarios. | Section 3 |
| 3 | Extend the model to other types of sport mega-events and evaluate their sustainable host selection. | Section 4 |
| 4 | Write a recommendation letter to the NFL. | Letter |

# 2 Part I: Identification and Modeling of Environmental Impact Factors for Sporting Events

## 2.1 Problem Analysis

Our overall approach to solving this problem is illustrated in Figure 1. Because energy consumption, greenhouse gas emissions, water usage, and waste generation involve numerous contributing factors, it is neither practical nor necessary to include all of them. Therefore, we established two guiding principles:

- Only the major contributing factors are retained, while secondary or negligible factors are omitted, because including all minor variables would introduce unnecessary complexity without significantly affecting the final results.

- All environmental footprints are limited to activities occurring within the host city, because impacts originating outside the citysuch as emissions from flights departing elsewhereare not directly caused by local event operations.

Following these two principles, we identified the key sub-factors for each environmental category and developed dedicated sub-models to quantify their respective contributions.

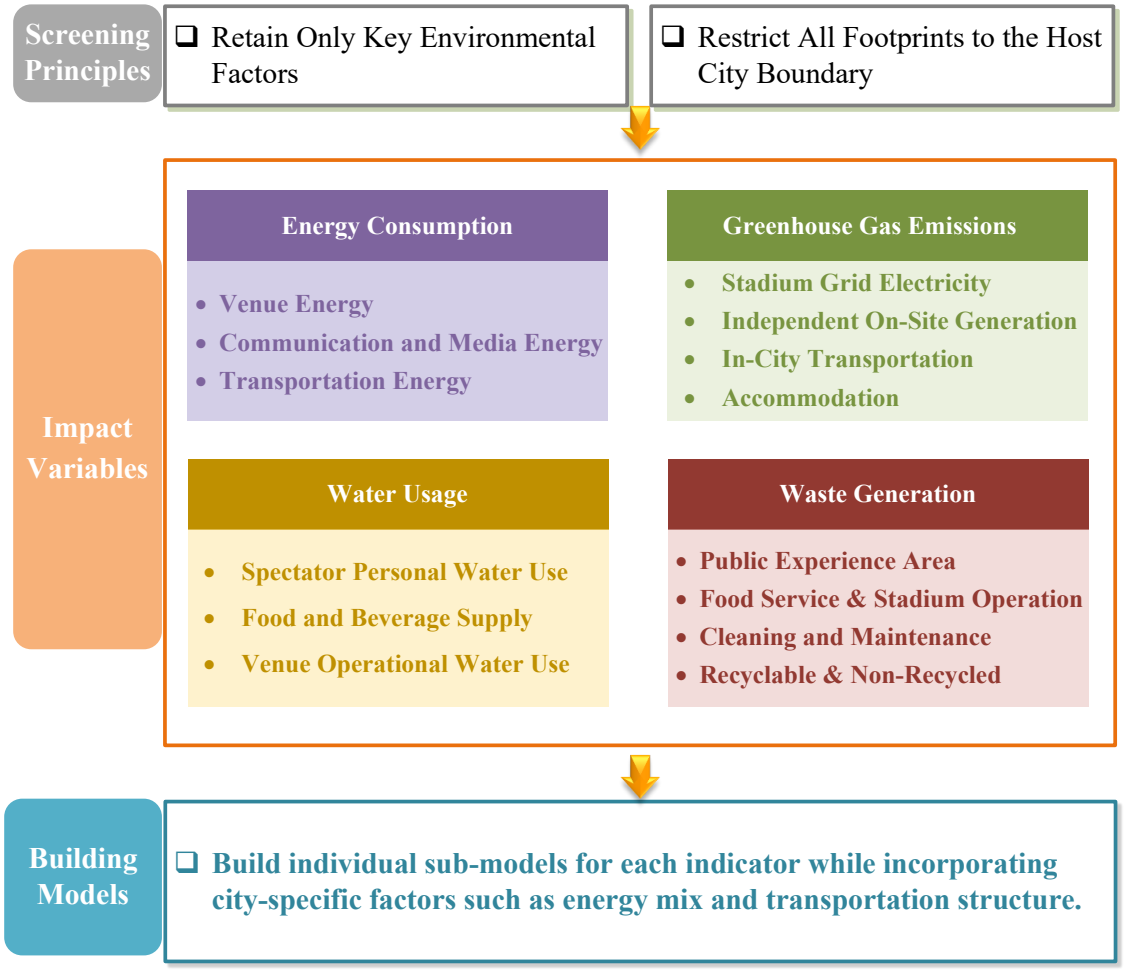



Figure 1: Framework of the Environmental Evaluation Model

## 2.2 Modeling: Energy Consumption

To clarify the composition of energy consumption, the total energy use is divided into three major parts:

$$E_{total} = E_{venue} + E_{broadcast} + E_{transport} \tag{1}$$

where $E_{venue}$ represents the energy consumption for venue operations, including lighting, HVAC systems, and basic equipment; $E_{broadcast}$ represents the energy consumption of communication and media systems, including signal transmission, data processing, and broadcasting system operations; and $E_{transport}$ represents the transportation energy consumption generated by the travel of spectators and staff.

**Venue Energy Consumption Model.** Venue energy consumption is the dominant part of energy use during events. Based on venue scale, audience size, and meteorological conditions, the venue energy consumption can be expressed as:

$$E_{venue} = \alpha_1 A + \alpha_2 N + \alpha_3 T_{avg} + \alpha_4 D \tag{2}$$

where $A$ is the venue building area ($m^2$), $N$ is the average number of spectators per day, $T_{avg}$ is the average temperature during the event ($^\circ C$), $D$ is the number of event days, and $\alpha_i$ are energy impact coefficients obtained through regression from historical event data.

**Communication and Media Energy Consumption Model.** The energy consumption of communication and media systems mainly depends on event scale and media coverage intensity. It can be modeled as:

$$E_{broadcast} = \beta_1 M + \beta_2 D \tag{3}$$

where $M$ represents the media intensity index, which can include the number of broadcast signals, online viewership, or quantified bandwidth demand; $D$ represents the number of event days; and $\beta_i$ are the statistical coefficients derived from historical event energy consumption data. This model enables the estimation of broadcasting energy use when detailed device-level power data are unavailable.

**Transportation Energy Consumption Model.** Transportation energy consumption mainly arises from the travel of spectators and staff. It can be estimated using macro-level statistical data as follows:

$$E_{transport} = \eta \times N_{att} \times D_{avg} \times e_{unit} \tag{4}$$

where $E_{transport}$ represents the total transportation energy consumption in kilowatt-hours, $N_{att}$ represents the total number of participants during the event, $D_{avg}$ represents the average one-way travel distance per person in kilometers, $e_{unit}$ represents the average energy consumption per kilometer per person in kilowatt-hours, and $\eta$ represents the round-trip coefficient.

## 2.3 Modeling: Greenhouse Gas Emissions

To restrict the analysis boundary to activities occurring within the host city, all intercity and air-travel emissions are excluded. The total greenhouse gas (GHG) footprint associated with the Super Bowl is expressed as

$$\text{GHG}_{total} = \text{GHG}_{grid} + \text{GHG}_{gen} + \text{GHG}_{transport,intra} + \text{GHG}_{lodging} \tag{5}$$

where $\text{GHG}_{grid}$ denotes emissions from grid-connected electricity use, $\text{GHG}_{gen}$ represents emissions from temporary on-site generation if used, $\text{GHG}_{transport,intra}$ quantifies mobility-related emissions within the host city, and $\text{GHG}_{lodging}$ captures emissions from hotel accommodation.

**Stadium Grid Electricity.** Electricity supplied by the regional grid for lighting, HVAC loads, broadcasting and temporary infrastructure is modeled as

$$\text{GHG}_{grid} = \sum_t E_t \, EF_{grid} \tag{6}$$

where $E_t$ is the stadium electricity demand during period $t$ (kWh), and $EF_{grid}$ is the average regional grid emission factor (kg $CO_2$e kWh$^{-1}$). To ensure reliable data availability, electricity demand is approximated using an activity-intensity formulation:

$$E_t \approx I_{act} \frac{N_{pax}}{\rho_{crowd}} L_{act} \tag{7}$$

where $I_{act}$ denotes the average emission intensity for stadium activities (kg $CO_2$e m$^{-2}$ day$^{-1}$), $N_{pax}$ the number of attendees, $\rho_{crowd}$ the average standing density (m$^{-2}$ person$^{-1}$), and $L_{act}$ the event duration (days).

**Independent On-Site Generation.** If temporary diesel generators support broadcasting or critical back-up loads, the associated emissions are

$$\text{GHG}_{gen} = \left( \sum_t E_{gen,t} \right) \frac{EF_{fuel}}{\eta_g} \tag{8}$$

where $E_{gen,t}$ is electricity produced by generators during period $t$, $EF_{fuel}$ is the fuel-specific emission factor (kg $CO_2$e kWh$^{-1}$ of fuel energy), and $\eta_g$ is the generator's conversion efficiency from fuel energy to electricity.

**In-City Transportation.** With intercity travel removed from the system boundary, only mobility within the metropolitan area is included. The total in-city transportation emissions are

$$\text{GHG}_{transport,intra} = N_{pax}\, L\, Freq\, \ell_{intra}\, e_{intra} \tag{9}$$

where $N_{pax}$ is the total number of attendees, $L$ the average length of stay (days), $Freq$ the average daily trip count per attendee, $\ell_{intra}$ the mean distance traveled per trip (km), and $e_{intra}$ the mode-weighted emission factor for urban mobility (kg $CO_2$e km$^{-1}$), incorporating public transit, ride-hail vehicles, shuttle services, and private cars.

**Accommodation.** Lodging emissions from hotel stays are modeled through the total number of room-nights:

$$RN = \sum_x \left( N_{pax}\, \text{Prop}_x \right) \frac{L\, \gamma_x}{x} \tag{10}$$

where $\text{Prop}_x$ is the share of attendee groups of size $x$, $\gamma_x$ the number of rooms required per group, and $L$ the length of stay. The resulting accommodation emissions are

$$\text{GHG}_{lodging} = RN\, IC_{rm}\, (1 - \rho_{green}) \tag{11}$$

where $IC_{rm}$ denotes the average hotel GHG emissions per room-night (kg $CO_2$e room$^{-1}$ night$^{-1}$) and $\rho_{green}$ the renewable share of hotel electricity supply.

## 2.4 Modeling: Water Usage

We study reconstructs the conventional model of "direct + indirect + operational" water use into a data-driven layered framework. All parameters in the model can be obtained from publicly available data sources such as municipal water-supply statistics, facility operation records, and event logistics reports. The total water consumption is divided into three components: personal water use $W_p$, virtual water use $W_v$, and operational facility water use $W_o$, as expressed by

$$W_{total} = W_p + W_v + W_o \tag{12}$$

**Spectator Personal Water Use.** Spectators personal water consumption mainly includes drinking, sanitation, and washing, which depend on attendance, event duration, and ambient temperature. Based on municipal daily per-capita water use and event characteristics, the spectator water-use model is formulated as

$$W_p = N_{att} \cdot D_{event} \cdot \overline{w}_{city} \left[1 + \alpha_T (T_{avg} - T_0)\right] \left[\beta_d r_d + \beta_s r_s + \beta_h r_h\right] \tag{13}$$

where $N_{att}$ represents the total number of attendees; $D_{event}$ denotes the number of event days; $\overline{w}_{city}$ refers to the average daily domestic water consumption per capita; $T_{avg}$ and $T_0$ are the mean temperature during the event and the reference temperature, respectively; $\alpha_T$ is a temperature-sensitivity coefficient; $r_d$, $r_s$, and $r_h$ are the proportional shares of drinking, sanitation, and hygiene activities; and $\beta_d$, $\beta_s$, and $\beta_h$ are the corresponding weighting coefficients that satisfy $\beta_d + \beta_s + \beta_h = 1$.

**Virtual Water Use in Food and Beverage Supply.** The indirect water consumption associated with food and beverage services represents the virtual water embodied in production, processing, and transportation. Using publicly accessible databases such as the Water Footprint Network (WFN), the total virtual water footprint can be expressed as

$$W_v = \sum_{i=1}^{n} (Q_i \cdot WF_i \cdot \eta_i) + \gamma_L \cdot L_{trans} \cdot WF_{fuel} \tag{14}$$

where $Q_i$ is the total consumption quantity of the $i$th food or beverage item, obtained from vendor or concession data; $WF_i$ is the blue-water footprint per unit of product $i$ ; $\eta_i$ denotes the regional water-scarcity correction factor derived from FAO or national statistics; $\gamma_L$ is the logistics weighting coefficient that accounts for transportation distance and refrigeration energy; $L_{trans}$ is the total transportation distance in kilometers; and $WF_{fuel}$ represents the water footprint of fuel consumption per unit of energy.

**Venue Operational Water Use.** Operational water consumption includes cleaning, cooling, and maintenance processes within the venue. Considering the coupling between energy and water use, this study introduces energy-to-water conversion factors to establish a more responsive and data-oriented model:

$$W_o = D_{event} \left[A_{serv} \cdot w_{clean} + \rho_{HVAC} \cdot E_{HVAC} + \rho_{maint} \cdot N_{staff}\right] (1 - \theta_{save}) \tag{15}$$

where $A_{serv}$ denotes the service area of the venue in square meters, as obtained from architectural or operation manuals; $w_{clean}$ is the cleaning water use per unit area per day; $E_{HVAC}$ is the

electricity consumed by HVAC systems during the event; $\rho_{HVAC}$ is the water demand per unit of HVAC electricity consumption; $\rho_{maint}$ represents the daily water consumption per maintenance staff member; $N_{staff}$ is the number of maintenance staff engaged during the event period; and $\theta_{save}$ is the water-saving ratio that reflects the effectiveness of conservation technologies .

## 2.5 Modeling: Waste Generation

To accurately describe the waste generation characteristics of large-scale events under accessible data conditions, the sources of waste are divided into four main parts: public experience areas, food service areas, stadium operation areas, and cleaning and maintenance areas. The total amount of waste generated during the event can be expressed as:

$$W_{total} = W_{exp} + W_{food} + W_{stadium} + W_{clean} - W_{recycle} \tag{16}$$

where $W_{exp}$ represents the waste from public experience areas, $W_{food}$ represents the waste from food services, $W_{stadium}$ represents the waste generated by spectators and stadium operations, $W_{clean}$ represents the waste from cleaning and maintenance processes, and $W_{recycle}$ represents the recyclable or recovered waste volume.

**Public Experience Area Waste.** Public experience areas mainly include promotional, interactive, and exhibition zones, where waste is primarily composed of paper materials, plastic packaging, and temporary decorative items. The waste generated in these areas can be estimated as:

$$W_{exp} = \alpha_1 N_{att} \left( \frac{A_{exp}}{A_{total}} \right)^{\beta_1} T_{event} \tag{17}$$

where $N_{att}$ denotes the total number of participants, $A_{exp}$ denotes the area of public experience zones ($m^2$), $A_{total}$ denotes the total venue area ($m^2$), $T_{event}$ denotes the event duration (h), $\alpha_1$ is the per capita waste generation coefficient (kg/personůh), and $\beta_1$ is the area elasticity coefficient (typically 0.91.1).

**Food Service Waste.** Food-related waste mainly consists of organic residues, food containers, and biodegradable packaging materials. Its generation depends on the scale of food services and recycling efficiency, which can be represented as:

$$W_{food} = \alpha_2 N_{meal}(1 - \eta_{food}) \tag{18}$$

where $N_{meal}$ represents the total number of meals provided (can be approximated by $N_{att} \times r_{meal}$), $r_{meal}$ represents the average number of meals per person, $\eta_{food}$ represents the composting or recovery rate of food waste, and $\alpha_2$ is the average waste generation per meal (kg/meal).

**Stadium Operation Waste.** During stadium operations, the main waste types include beverage containers, paper cups, and plastic packaging. The total amount can be estimated as:

$$W_{stadium} = \alpha_3 N_{att} \gamma_{seat}(1 - \eta_{stadium}) \tag{19}$$

where $\gamma_{seat}$ denotes the seat occupancy rate (ratio of occupied to total seats), $\eta_{stadium}$ denotes the recovery or sorting rate within the stadium, and $\alpha_3$ represents the average waste generation per spectator (kg/person).

**Cleaning and Maintenance Waste.** Cleaning and maintenance waste mainly comes from ground litter, packaging residues, and temporary facility dismantling materials. It can be expressed as:

$$W_{clean} = \alpha_4 A_{clean} \delta_{clean} \tag{20}$$

where $A_{clean}$ denotes the total cleaning area ($m^2$), $\delta_{clean}$ denotes the waste generation intensity per unit area ($kg/m^2$), and $\alpha_4$ is the cleaning coefficient reflecting surface contamination level and crowd density.

**Recyclable Waste Volume.** To represent the contribution of recycling and resource recovery, the average recovery rate is defined as $\eta_{avg}$, and the recyclable waste volume is expressed as:

$$W_{recycle} = \eta_{avg}(W_{exp} + W_{food} + W_{stadium} + W_{clean}) \tag{21}$$

where $\eta_{avg}$ represents the overall recycling efficiency during the event.

**Effective Non-Recycled Waste.** By combining the above components, the effective non-recycled waste during the event can be written as:

$$W_{NR} = (1 - \eta_{avg})[\alpha_1 N_{att}(\frac{A_{exp}}{A_{total}})^{\beta_1} T_{event} + \alpha_2 N_{meal} + \alpha_3 N_{att} \gamma_{seat} + \alpha_4 A_{clean} \delta_{clean}] \tag{22}$$

## 2.6 Model Results

Using Super Bowl LIV as the baseline scenario and inputting it into the environmental indicator model yields the corresponding breakdown of environmental impacts.

Figure 2 decomposes the primary sources of environmental burden associated with the event. The results show that energy consumption is dominated by transportation, accounting for nearly 80%, far exceeding stadium operations and broadcasting. Carbon emissions are mainly driven by accommodation and transportation, together contributing more than 85%, indicating that spectator mobility and lodging patterns largely determine the overall carbon footprint. In terms of water use, the implicit water footprint linked to digital activities represents the largest share, approximately 60%. Waste generation arises primarily from spectator areas and food services, concentrated in high-turnover disposable materials.

The modeled indicator distribution shows that most of the Super Bowls environmental burden stems from off-venue activities. Transportation contributes nearly 80% of energy use, while accommodation and travel generate over 85% of carbon emissions, underscoring spectator mobility as the main driver. About 60% of water stress comes from indirect water use, and waste is concentrated in spectator and food-service areas. Overall, peripheral activitiesnot the game itselfaccount for the majority of environmental impacts.

# 3 Part II: Comprehensive Evaluation Model Based on the Entropy Weight Method and TOPSIS

## 3.1 Problem Analysis

The problem requires selecting the most environmentally sustainable host city for the Super Bowl from a set of candidate locations. This task is inherently a multi-criteria decision-making

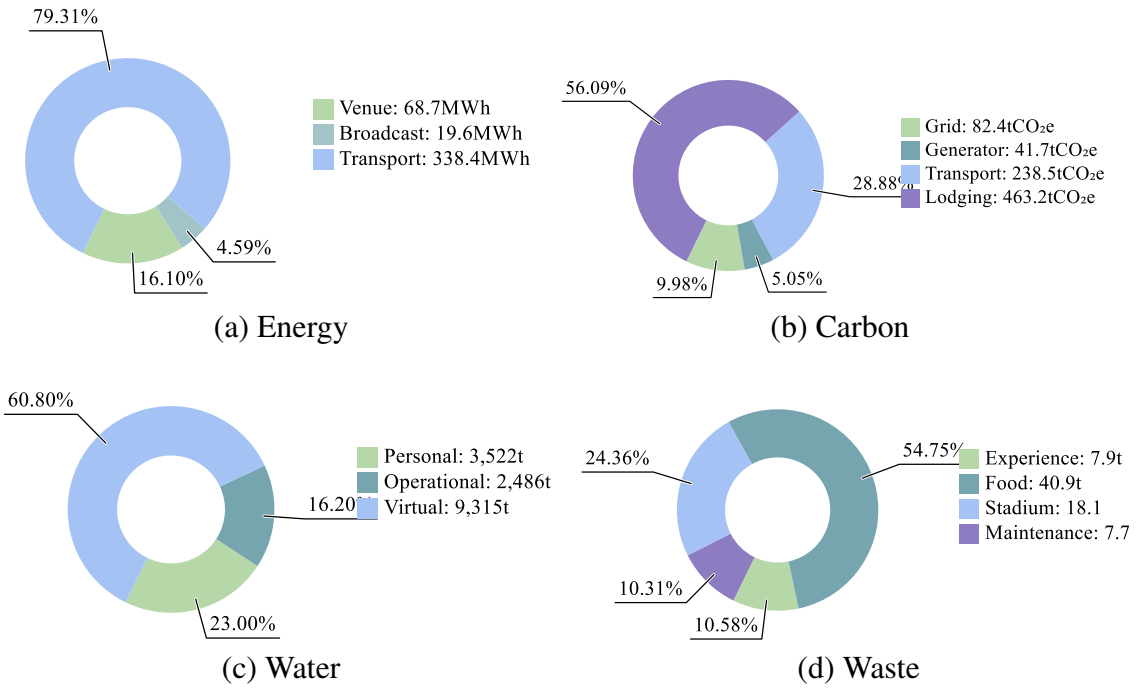



Figure 2: Source-Based Breakdown of Environmental Impacts for Super Bowl LIX

(MCDM) problem, as each city must be evaluated across multiple heterogeneous indicators such as energy consumption, water use, waste generation, and pollutant emissions.

To address this, we construct a comprehensive evaluation model that integrates the entropy weight method and the TOPSIS (Technique for Order Preference by Similarity to Ideal Solution) approach [2]. First, the environmental indicators for each city are obtained from the four models developed in Section II. These indicators are then normalized to ensure comparability. The entropy weight method is applied to compute objective, data-driven weights for all indicators. Finally, the weighted dataset is assessed using the TOPSIS method to produce a final ranking of cities. The overall modeling workflow is shown in Fig. 3.

## 3.2 Entropy Weight Method: Deriving Objective Indicator Weights

The entropy weight method is an objective weighting technique used to quantify the relative importance of indicators according to the degree of variation they exhibit across evaluation objects. Indicators with greater variability provide more information and are therefore assigned higher weights. Conversely, indicators with lower variability are deemed less informative and thus receive smaller weights.

Let $X = (x_{ij})_{m\times n}$ denote the original data matrix, where $i = 1, \ldots, m$ indexes the candidate cities and $j = 1, \ldots, n$ indexes the environmental indicators. The entropy weight method proceeds as follows.

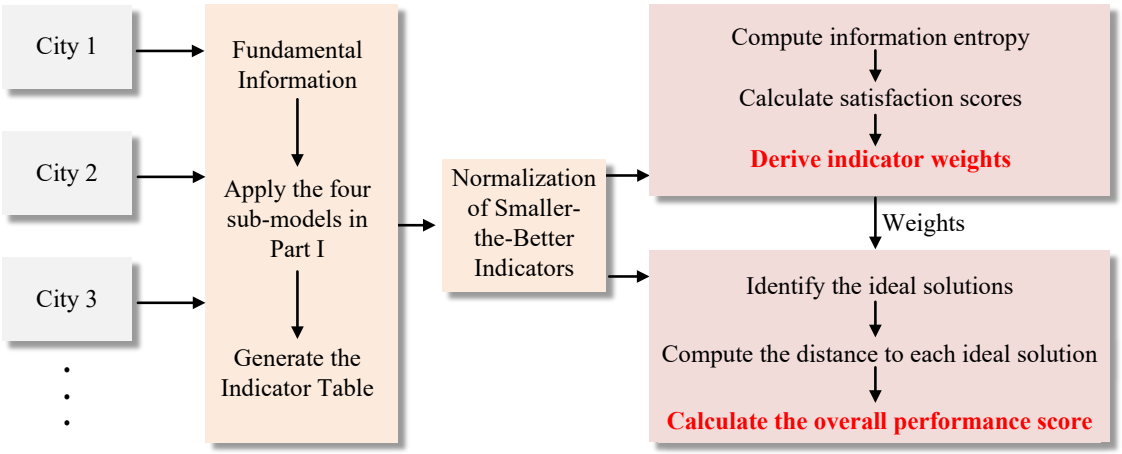



Figure 3: Overall workflow of the entropy–TOPSIS comprehensive evaluation model.

**Step 1:** Normalize the data. To eliminate dimensional inconsistencies, the indicators are normalized using min–max scaling:

$$z_{ij} = \frac{x_{ij} - \min_i x_{ij}}{\max_i x_{ij} - \min_i x_{ij}}. \tag{23}$$

**Step 2:** Compute the proportion matrix. The contribution of city $i$ to indicator $j$ is given by:

$$p_{ij} = \frac{z_{ij}}{\sum_{i=1}^{m} z_{ij}}, \qquad \text{if } \sum_{i=1}^{m} z_{ij} = 0, \text{ then } p_{ij} = \frac{1}{m}. \tag{24}$$

**Step 3:** Calculate the entropy of each indicator.

$$e_j = -k \sum_{i=1}^{m} p_{ij} \ln p_{ij}, \qquad k = \frac{1}{\ln m}. \tag{25}$$

**Step 4:** Compute the degree of divergence and normalized weights.

$$d_j = 1 - e_j, \qquad w_j = \frac{d_j}{\sum_{j=1}^{n} d_j}. \tag{26}$$

The resulting vector $w = (w_1, \ldots, w_n)$ represents the objective information-based importance of each indicator, forming the foundation for the subsequent TOPSIS analysis.

## 3.3 TOPSIS Method: Ranking Cities Based on Sustainability Performance

TOPSIS is a classical MCDM method that evaluates each alternative according to its relative distance to a positive ideal solution (best values of all indicators) and a negative ideal solution (worst values). A city is considered more sustainable if it is closer to the ideal point and farther from the anti-ideal point. Given the decision matrix $X = (x_{ij})$ and entropy-derived weights $w = (w_1, \ldots, w_n)$, TOPSIS proceeds as follows.

**Step 1:** Vector normalization.

$$r_{ij} = \frac{x_{ij}}{\sqrt{\sum_{i=1}^{m} x_{ij}^2}}. \tag{27}$$

**Step 2:** Construct the weighted normalized matrix.

$$v_{ij} = w_j r_{ij}. \tag{28}$$

**Step 3:** Determine the ideal and anti-ideal solutions.

$$v_j^+ = \max_i v_{ij}, \qquad v_j^- = \min_i v_{ij}. \tag{29}$$

**Step 4:** Compute the Euclidean distances to the ideal solutions.

$$S_i^+ = \sqrt{\sum_{j=1}^{n} (v_{ij} - v_j^+)^2}, \qquad S_i^- = \sqrt{\sum_{j=1}^{n} (v_{ij} - v_j^-)^2}. \tag{30}$$

**Step 5:** Calculate the relative closeness coefficient.

$$C_i = \frac{S_i^-}{S_i^+ + S_i^-}, \qquad 0 \le C_i \le 1. \tag{31}$$

**Step 6:** Rank alternatives and identify the optimal city. A larger closeness coefficient $C_i$ indicates stronger overall environmental sustainability. Thus, all candidate cities are ranked in descending order of $C_i$, and the city with the highest value of $C_i$ is selected as the most environmentally sustainable host location.

## 3.4 Model Results

### 3.4.1 Case I: Selection from Cities with Previous Super Bowl Hosting Experience

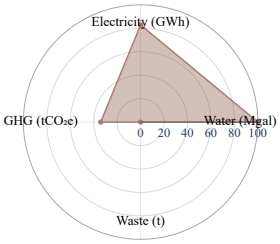
After organizing NFL operational data [3] together with publicly available energy [4] and water consumption [4] statistics, and applying our Python-based environmental assessment model alongside the entropy-weight and TOPSIS methods, we obtained the raw environmental indicators for each city as shown in Table 1. The data were then normalizedwhere values closer to 1 indicate better performanceas presented in Table 2. Corresponding radar charts are provided in Figure 4.

*Notation:* $E$ denotes electricity consumption (GWh); $W$ denotes water consumption (million gallons); $M$ denotes waste mass generated (tons); and $G$ represents greenhouse gas emissions measured in metric tons of $CO_2$ equivalent ($tCO_2e$).

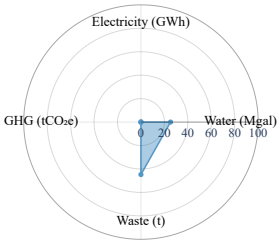
Table 1 reports the raw resource consumption of the six potential Super Bowl host cities across four indicators: energy use (E), water consumption (W), solid waste generation (M), and greenhouse gas emissions (G). The values vary substantially across cities, reflecting the influence of stadium scale, local energy structure, and operational practices on baseline environmental burdens.

Table 1: Environmental Resource Profile of Candidate Super Bowl Host Stadiums

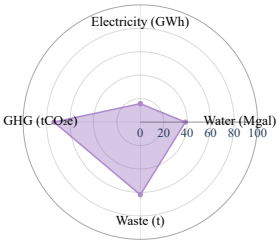
| City | Stadium | $E$ (MWh) | $W$ (t) | $M$ (t) | $G$ (t $CO_2$e) |
|---|---|---|---|---|---|
| Glendale, AZ | State Farm Stadium | 492.3 | 16280.4 | 82.1 | 910.7 |
| Las Vegas, NV | Allegiant Stadium | 445.8 | 17120.7 | 74.9 | 892.4 |
| New Orleans, LA | Caesars Superdome | 426.7 | 15323.0 | 75.2 | 825.8 |
| Santa Clara, CA | Levi's Stadium | 402.5 | 14510.2 | 59.8 | 880.1 |
| Inglewood, CA | SoFi Stadium | 478.2 | 15840.5 | 71.0 | 842.9 |
| Atlanta, GA | Mercedes-Benz Stadium | 417.9 | 13790.1 | 89.3 | 868.6 |



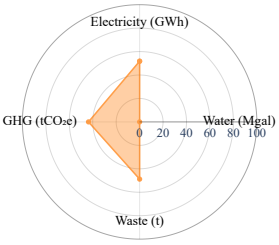

(a) Atlanta GA



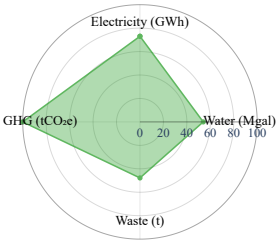

(b) Glendale AZ



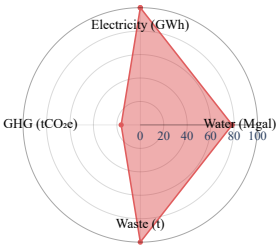

(c) Inglewood CA




(d) Las Vegas NV




(e) New Orleans LA




(f) Santa Clara CA

Figure 4: Environmental Performance Radar Charts of Candidate Super Bowl Host Cities

From the dimensionless scores in Table 2 and the radar charts in Figure 4, it becomes evident that Santa Clara and New Orleans hold broad advantages across all sub-indicators.

The entropy-based indicator weights are shown in Table 3, reveal that GHG receives the highest weight ($w_W = 0.2918$), reflecting its relatively low entropy value and strong capacity to differentiate performance across cities. Electricity also attains a substantial weight ($w_E = 0.2605$), indicating meaningful variability among the six locations.

The TOPSIS evaluation results in Table 4 indicate that New Orleans is the closest to the positive ideal solution, driven by its strong performance in electricity use, water consumption, and carbon emissions. Santa Clara ranks second, though its relatively higher carbon emissions reduce its overall score. As shown in Figure 5, Atlanta, Inglewood, Las Vegas, and Glendale lie much farther from the positive ideal solution and closer to the negative ideal, appearing in the lower-right region of the plot. Consequently, their overall scores are notably lower compared with New Orleans and Santa Clara.

Table 2: Normalized Environmental Indicator Matrix for Candidate Host Cities

| City | Stadium | $E$ | $W$ | $M$ | $G$ |
|---|---|---|---|---|---|
| Glendale, AZ | State Farm Stadium | 0.0000 | 0.2523 | 0.4475 | 0.0000 |
| Las Vegas, NV | Allegiant Stadium | 0.5178 | 0.0000 | 0.4881 | 0.4361 |
| New Orleans, LA | Caesars Superdome | 0.7305 | 0.5398 | 0.4780 | 1.0000 |
| Santa Clara, CA | Levi's Stadium | 1.0000 | 0.7838 | 1.0000 | 0.1633 |
| Inglewood, CA | SoFi Stadium | 0.1570 | 0.3844 | 0.6203 | 0.7365 |
| Atlanta, GA | Mercedes-Benz Stadium | 0.8285 | 1.0000 | 0.0000 | 0.3405 |

Table 3: Entropy, Redundancy, and Weight of Each Indicator

| Indicator | Entropy $e_j$ | Redundancy $d_j$ | Weight $w_j$ |
|---|---|---|---|
| Electricity (E) | 0.8305 | 0.1695 | 0.2605 |
| Water (W) | 0.8392 | 0.1608 | 0.2470 |
| Waste (M) | 0.8694 | 0.1306 | 0.2007 |
| GHG (G) | 0.8101 | 0.1899 | 0.2918 |

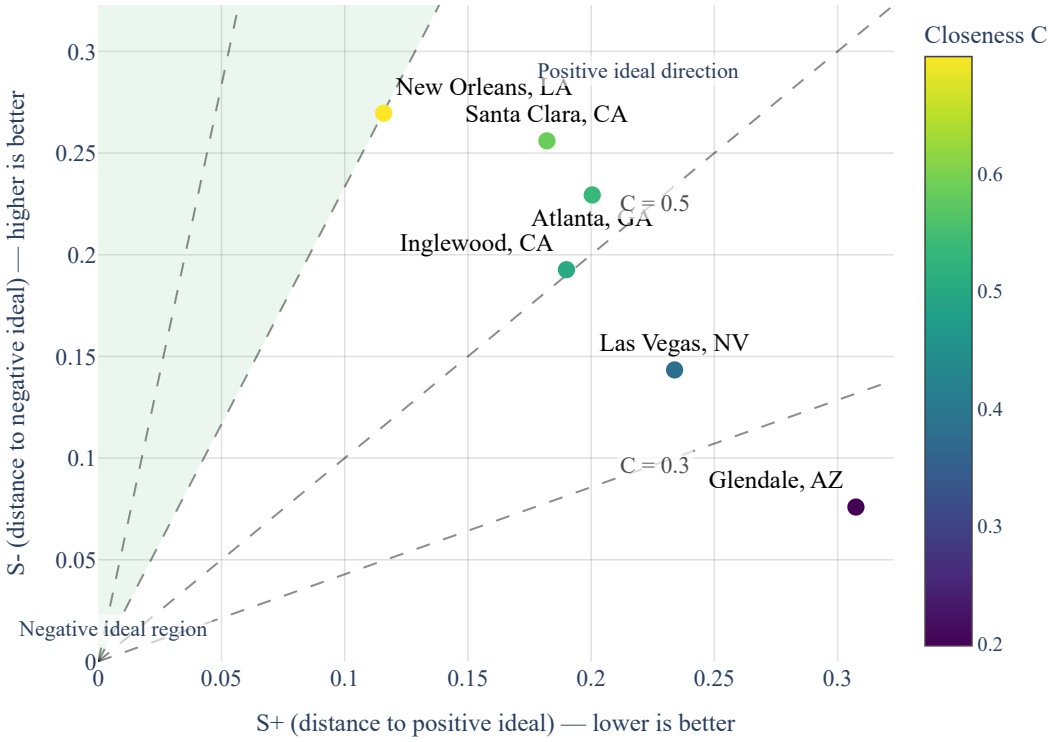



Figure 5: Visualization of TOPSIS Results for Candidate Host Cities

In summary, the entropy-TOPSIS analysis identifies New Orleans and Santa Clara as the strongest candidates for sustainable Super Bowl hosting, with New Orleans showing the smallest distance to the ideal solution. Their advantages in electricity use, water consumption, and GHG emissions clearly distinguish them from the other four cities, which cluster closer to the negative

Table 4: TOPSIS Results for Candidate Host Cities

| City | $S^+$ | $S^-$ | Closeness $C$ | Rank |
|---|---|---|---|---|
| Glendale, AZ | 0.307438 | 0.075927 | 0.198055 | 6 |
| Las Vegas, NV | 0.233857 | 0.143356 | 0.380040 | 5 |
| New Orleans, LA | 0.115827 | 0.269634 | 0.699510 | 1 |
| Santa Clara, CA | 0.182039 | 0.256017 | 0.584439 | 2 |
| Inglewood, CA | 0.189976 | 0.192648 | 0.503492 | 4 |
| Atlanta, GA | 0.200434 | 0.229397 | 0.533691 | 3 |

ideal. The results demonstrate that substantial differences in stadium efficiency and local resource structures critically shape the environmental competitiveness of potential host cities.

### 3.4.2 Case II: Selection from Cities without Prior Super Bowl Hosting Experience

For NFL home stadiums that have not previously hosted a Super Bowl, we randomly selected four cities and their venues: Houston (NRG Stadium), Minneapolis (U.S. Bank Stadium), Indianapolis (Lucas Oil Stadium), and Charlotte (Bank of America Stadium). By collecting data on stadium characteristics [6] and local grid structures, we obtained the raw model indicators summarized in Table 5.Overall, the environmental burdens among the four cities exhibit relatively limited variation.

Table 5: Estimated Resource Consumption and Emissions for New Super Bowl Host Cities

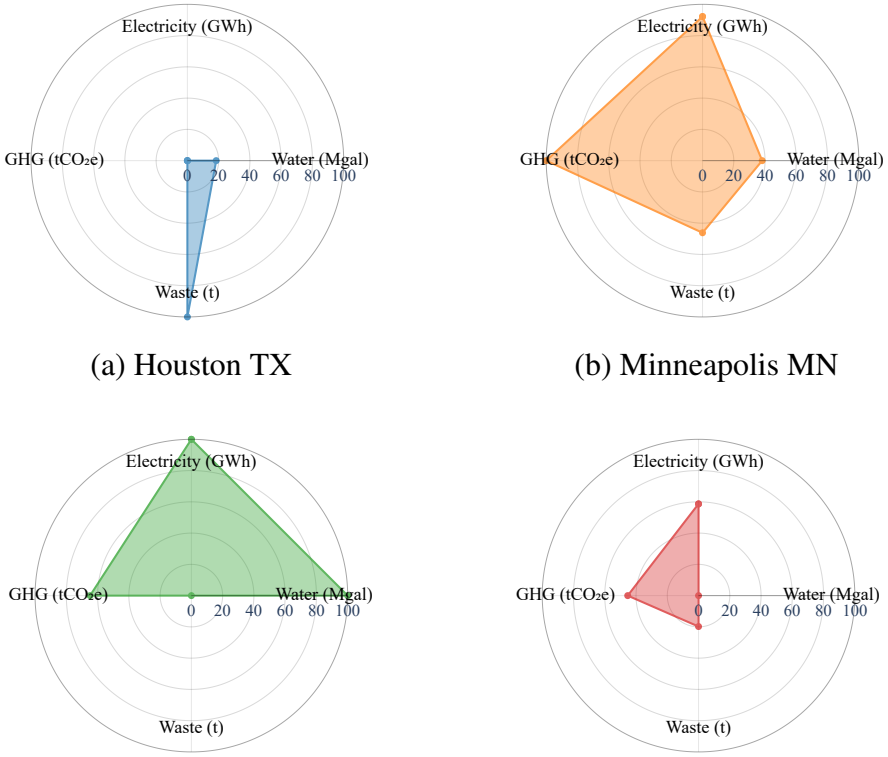
| City | Stadium | $E$ (MWh) | $W$ (t) | $M$ (t) | $G$ (t $CO_2$e) |
|---|---|---|---|---|---|
| Houston, TX | NRG Stadium | 512.4 | 15846 | 64.5 | 832.7 |
| Minneapolis, MN | U.S. Bank Stadium | 458.2 | 15693 | 74.3 | 728.1 |
| Indianapolis, IN | Lucas Oil Stadium | 453.6 | 15228 | 82.7 | 764.9 |
| Charlotte, NC | Bank of America Stadium | 477.9 | 15980 | 79.1 | 785.3 |

Table 6: Normalized Environmental Indicator Matrix for New Super Bowl Host Cities

| City | Stadium | $E$ | $W$ | $M$ | $G$ |
|---|---|---|---|---|---|
| Houston, TX | NRG Stadium | 0.0000 | 0.1842 | 1.0000 | 0.0000 |
| Minneapolis, MN | U.S. Bank Stadium | 0.9218 | 0.3816 | 0.4615 | 1.0000 |
| Indianapolis, IN | Lucas Oil Stadium | 1.0000 | 1.0000 | 0.0000 | 0.6482 |
| Charlotte, NC | Bank of America Stadium | 0.5867 | 0.0000 | 0.1978 | 0.4532 |

Table 6 and Figure 6 present the normalized indicator values and corresponding radar charts for the four cities. The results show clear differentiation in strengths and weaknesses across indicators. Minneapolis performs well in both greenhouse gas emissions and energy use, displaying a more balanced multi-dimensional profile, whereas the other three cities exhibit more pronounced shortcomings in specific indicators.

Table 7 reports the information entropy, redundancy, and final weights of the four environmental indicators derived through the entropy-weight method. The TOPSIS results, shown in Table 8, indicate that Indianapolis ranks first, benefiting from strong performance in the high-weight energy




(a) Houston TX (b) Minneapolis MN

(c) Indianapolis IN (d) Charlotte NC

Figure 6: Environmental Performance Radar Charts of New Super Bowl Host Cities

Table 7: Entropy, Redundancy, and Weights of Environmental Indicators

| Indicator | Entropy $e_j$ | Redundancy $d_j$ | Weight $w_j$ |
|---|---|---|---|
| Electricity (E) | 0.77497 | 0.22503 | 0.19174 |
| Water (W) | 0.63637 | 0.36363 | 0.30983 |
| Waste (M) | 0.65978 | 0.34022 | 0.28989 |
| GHG (G) | 0.75526 | 0.24474 | 0.20854 |

and water dimensions. Minneapolis follows with a relatively balanced profile across all indicators. In contrast, Houston and Charlotte deviate more substantially from the ideal solution, as illustrated in Figure 7, reflecting notable weaknesses in specific indicators and resulting in lower overall evaluations.

In summary, applying the entropyTOPSIS framework to the four newly selected cities highlights Indianapolis as the most sustainable candidate, driven by its strong performance in the high-weight energy and water indicators. Minneapolis follows closely with a balanced overall profile. Houston and Charlotte show clear weaknesses across multiple dimensions, placing them farther from the ideal solution and resulting in lower sustainability scores.

Table 8: TOPSIS Results for New Super Bowl Host City Candidates

| City | Stadium | $S^+$ | $S^-$ | Closeness $C$ | Rank |
|---|---|---|---|---|---|
| Houston, TX | NRG Stadium | 0.312528 | 0.264338 | 0.458231 | 3 |
| Minneapolis, MN | U.S. Bank Stadium | 0.225137 | 0.259110 | 0.535078 | 2 |
| Indianapolis, IN | Lucas Oil Stadium | 0.265375 | 0.330732 | 0.554819 | 1 |
| Charlotte, NC | Bank of America Stadium | 0.368017 | 0.117850 | 0.242556 | 4 |

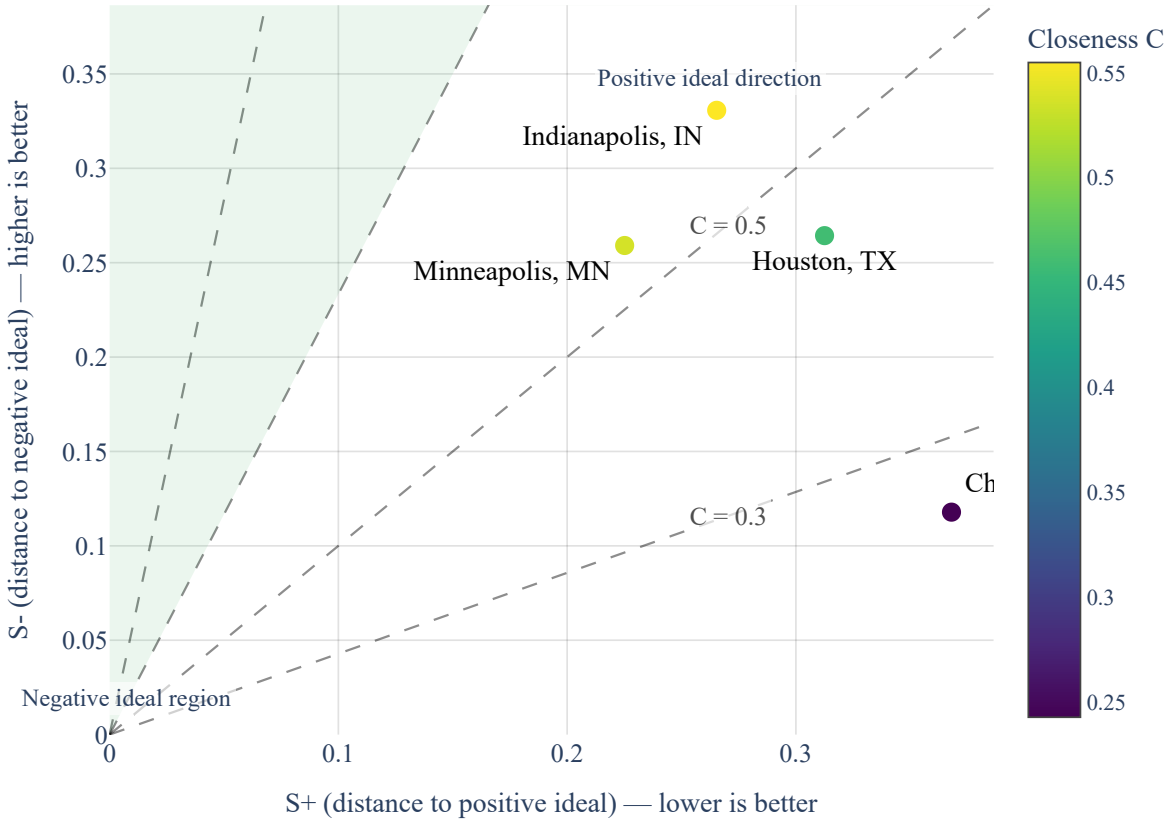



Figure 7: Visualization of TOPSIS Results for New Super Bowl Host Cities

# 4 Part III: Extended Modeling Framework for Single-Sport Championship Tournaments

## 4.1 Structural Differences Between the Super Bowl and Single-Sport Championship Tournaments

The sustainability assessment framework developed earlier in this study was tailored to the specific characteristics of the Super Bowl, an event defined by a "single-city, single-game, short-duration" structure. Under this structure, environmental impacts—including energy use, water consumption, waste generation, and greenhouse gas emissions—are concentrated within a one-week window and peak on game day. All resource flows and emissions occur within a fixed urban

boundary, enabling the environmental footprint to be evaluated as a set of localized, event-specific components.

In contrast, single-sport championship tournaments such as the Formula 1 Grand Prix [7], ATP/WTA tennis tours [8], or cycling stage races differ fundamentally from the Super Bowl in both temporal and spatial dimensions. These tournaments typically unfold across multiple weeks, span several host cities, and follow a sequential multi-stage competitive format. Teams, staff members, media crews, and spectators move from one host location to another, generating substantial intercity transportation activity. Furthermore, operational loads (energy, water, and waste) shift geographically as the tournament progresses, and each stage may differ significantly in venue size, attendance, and resource intensity.

To clearly articulate the structural differences between the Super Bowl and multi-stage single-sport tournaments, Table 9 summarizes the key distinctions relevant to sustainability modeling.

Table 9: Differences Between the Super Bowl and Single-Sport Championship Tournaments

| Dimension | Super Bowl | Single-Sport Tournament |
|---|---|---|
| Time span | $\sim$1 week | Multi-week (4–40 weeks) |
| Number of venues | 1 | 5–20 venues |
| Spatial distribution | Single-city | Multi-city / multi-country |
| Event structure | Single game | Sequential stages or rounds |
| Spectator mobility | Mostly intra-city | Intercity and international mobility |
| Team/media logistics | Stationary | Continuous relocation across stages |
| Operational loads | Localized | Distributed and cumulative |
| System boundary | Fixed | Dynamic and stage-dependent |

These structural differences imply that the original Super Bowl model cannot be directly applied to multi-stage tournaments: system boundaries become dynamic, transport emissions grow in importance, and each stage contributes differently to total environmental impact. Therefore, each of the four sub-models—energy, greenhouse gases, water use, and waste generation—must be expanded to incorporate stage-wise accumulation and spatial heterogeneity. The following subsections present the extended formulations.

## 4.2 Model Extension for Single-Sport Championship Tournaments

### 4.2.1 Extension of the Energy Consumption Model

In the energy consumption model, total energy use comprises $E_{venue}$, $E_{broadcast}$, and $E_{transport}$. For multi-stage single-sport tournaments, the first two components can be modeled by computing the energy consumption of each stage independently and aggregating them across the full tournament.

$$E_{venue} = \sum_s E_{venue,s} = \sum_s (\alpha_1 A_s + \alpha_2 N_s + \alpha_3 T_{avg,s} + \alpha_4 D_s) \tag{32}$$

$$E_{broadcast} = \sum_s E_{broadcast,s} = \beta_1 M_s + \beta_2 D_s \tag{33}$$

where the subscript $s$ indicates the parameters associated with different stages.

In addition to accounting for multiple stages, $E_{transport}$ must also incorporate the additional energy required for team and equipment relocation between host cities. This can be expressed as follows:

$$\begin{aligned} E_{transport} &= \sum_{s} (E_{transport,s} + E_{move,s->s+1}) \\ &= \sum_{s} (\eta_s \times N_{att,s} \times D_{avg,s} \times e_{unit} + N_{staff} \times D_{s->s+1} \times e_{unit}) \end{aligned} \tag{34}$$

where $N_{staff}$ denotes the number of staff members, and $D_{s \to s+1}$ represents the distance between the host locations of stages $s$ and $s+1$.

### 4.2.2 Extension of the Greenhouse Gas Emissions Model

Similar to the extension made for $E_{venue}$, all four components of $GHG_{total}$ can be formulated as the summation of greenhouse gas emissions across multiple stages. The specific equations for each component are omitted here for brevity. It is important to note, however, that due to differences in grid composition across host cities, the parameter values for each stage may require adjustment.

$$\begin{aligned} GHG_{total} &= \sum_{s} GHG_{total,s} \\ &= \sum_{s} (GHG_{grid,s} + GHG_{gen,s} + GHG_{transport,intra,s} + GHG_{lodging,s}) \end{aligned} \tag{35}$$

### 4.2.3 Extension of the Water Usage Model

In the water resource model, total water use comprises personal water consumption $W_p$, virtual water $W_v$, and venue operational water use $W_o$. For multi-stage single-sport tournaments, the water use of each stage can be computed separately and then aggregated. In addition, such tournaments typically require post-event field maintenance after each competition day, necessitating an additional component of field maintenance water use, denoted as $W_{field}$.

$$\begin{aligned} W_{total} &= \sum_{s} W_{total,s} \\ &= \sum_{s} (W_{p,s} + W_{v,s} + W_{o,s} + W_{field,s}) \\ &= \sum_{s} \big(W_{p,s} + W_{v,s} + W_{o,s} \\ &\quad + A_{field,s} \times w_{field} \times D_s\big) \end{aligned} \tag{36}$$

where $s$ denotes the stage index of the tournament, $A_{field,s}$ represents the surface area of the turf or competition field, and $w_{field}$ denotes the water-use intensity per unit area for field maintenance. By summing water consumption across all stages and venues, the above formulation captures the temporally and spatially accumulated water use of a single-sport championship tournament.

### 4.2.4 Extension of the Waste Generation Model

In the waste-generation model, the Super Bowl framework includes only a single-stage waste production component, with deductions made through the recyclable waste volume $W_{\text{recycle}}$. For multi-stage single-sport tournaments, waste from all sources must be aggregated across stages, and variations in recycling rates among host cities must be accounted for. Accordingly, the model should be reformulated in a stage-wise summation structure.

$$\begin{aligned} W_{total} &= \sum_{s} W_{total,s} \\ &= \sum_{s} (W_{\text{exp},s} + W_{food,s} + W_{stadium,s} + W_{clean,s} - W_{recycle,s}) \end{aligned} \tag{37}$$

## 4.3 Model Application

Among single-sport championship tournaments, badminton championships stand out as one of the most representative events for assessing host-city sustainability, due to their highly standardized competition format and their multi-city scheduling structure. Taking the BWF World Tour as an example, each season typically consists of consecutively hosted events across multiple cities in Asia, Europe, and North America, forming a continuous multi-city event chain. Compared with the single-location, short-duration Super Bowl, sustainability assessment for the World Tour is considerably more complex.

Although the World Tour maintains a relatively stable hierarchical event structure, the selection of host cities varies by season and is influenced by venue conditions, regional policies, and other factors. Different seasons exhibit notable differences in their spatial configuration of host cities, resulting in several feasible city clusters grounded in actual operational patterns. To operationalize the extended multi-stage sustainability model, we selects multiple representative sets of host-city combinations from recent World Tour schedules as simplified event-operation scenarios, as shown in Table 10 [9].

Table 10: Four Representative City Combinations of BWF World Tour (Top-Tier Events)

| Scheme | Year | Host Cities (Top-Tier Events) |
|---|---|---|
| A | 2021 | Jakarta, Kuala Lumpur, Changzhou, Odense, Singapore, Tokyo |
| B | 2020 | Jakarta, Kuala Lumpur, Birmingham, Odense, Singapore, Tokyo |
| C | 2019 | Bangkok, Jakarta, Birmingham, Odense, Tokyo, Singapore |
| D | 2018 | Jakarta, Birmingham, Wuhan, Odense, Fuzhou, Tokyo |

First, we calculated the environmental indicators for all cities involved in the four proposed schemes, with the raw results shown in Table 11. Following the sequence of each scheme, we then computed the inter-city transition costs and added them to the corresponding city-level impacts to obtain the complete environmental indicators for each BWF World Tour scheme, as presented in Table 12.

Table 11 shows that while resource consumption varies across BWF host cities, all indicators remain within a similar order of magnitude. Tokyo and Jakarta exhibit relatively high energy,

Table 11: Estimated Environmental Indicators for BWF World Tour Top-Tier Host Cities (Per Tournament Stop)

| City | Energy (MWh) | Water (t) | Waste (t) | Emissions (t $CO_2$e) |
|---|---|---|---|---|
| Jakarta | 263.4 | 5,874 | 43.1 | 426.0 |
| Kuala Lumpur | 239.7 | 5,536 | 38.6 | 384.6 |
| Changzhou | 229.5 | 5,421 | 40.2 | 405.5 |
| Odense | 214.1 | 4,927 | 34.3 | 318.2 |
| Singapore | 262.4 | 6,110 | 42.1 | 401.7 |
| Tokyo | 276.9 | 6,283 | 44.8 | 412.4 |
| Birmingham | 224.0 | 5,277 | 36.7 | 347.7 |
| Bangkok | 248.5 | 5,839 | 41.5 | 436.9 |
| Wuhan | 246.3 | 5,492 | 40.4 | 421.2 |
| Fuzhou | 233.1 | 5,383 | 37.2 | 414.6 |

Table 12: Aggregated Environmental Indicators for Four Representative BWF World Tour City Combinations

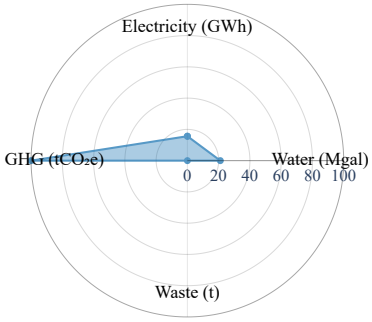
| Scheme | Energy (MWh) | Water (t) | Waste (t) | Emissions (t $CO_2$e) |
|---|---|---|---|---|
| A | 1,485.7 | 34,154 | 241.6 | 2,275.5 |
| B | 1,479.4 | 33,553 | 238.2 | 2,330.2 |
| C | 1,490.8 | 34,318 | 233.7 | 2,325.9 |
| D | 1,458.1 | 33,825 | 236.5 | 2,315.6 |

water, and waste levels due to their large urban scale, whereas Odense and Birmingham show lower values, reflecting the potential advantages of smaller cities. After incorporating inter-city transition costs, the total environmental impacts of the four tour schemes are obtained in Table 12, Figure 8. Overall differences are modest, though Scheme D has the lowest energy and water use, while Schemes B and C show slightly higher carbon emissions due to longer travel distances. This demonstrates that environmental performance depends on both city-level characteristics and the spatial arrangement of the tour.

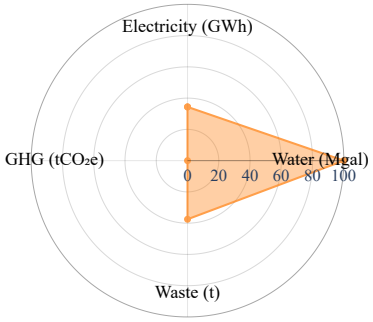
Table 13: Normalized Environmental Indicator Matrix for BWF World Tour Schemes

| Scheme | $E$ | $W$ | $M$ | $G$ |
|---|---|---|---|---|
| A | 0.1563 | 0.2105 | 0.0000 | 1.0000 |
| B | 0.3438 | 1.0000 | 0.3750 | 0.0000 |
| C | 0.0000 | 0.0000 | 1.0000 | 0.0909 |
| D | 1.0000 | 0.6447 | 0.6250 | 0.2727 |

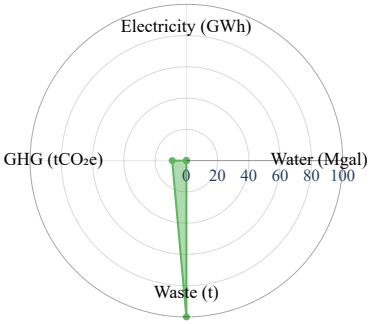
After normalization, the environmental indicators of each scheme are presented in Table 8. The entropy-based weights, shown in Table 14, indicate that greenhouse gas emissions and energy consumption carry the highest weights, making them the primary factors distinguishing the sustainability performance of the four schemes.



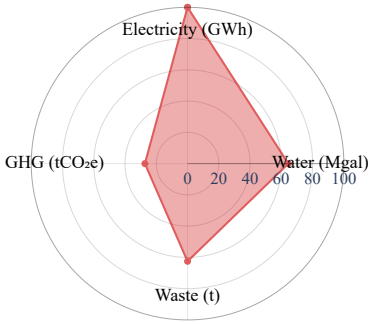

(a) Scheme A




(b) Scheme B




(c) Scheme C




(d) Scheme D

Figure 8: environmental Performance Radar Charts of BWF City-Combination Schemes

Table 14: Entropy, Redundancy, and Weights for BWF World Tour Schemes

| Indicator | Entropy $e_j$ | Redundancy $d_j$ | Weight $w_j$ |
|---|---|---|---|
| Energy (E) | 0.60849 | 0.39151 | 0.27131 |
| Water (W) | 0.68338 | 0.31662 | 0.21941 |
| Waste (M) | 0.73861 | 0.26139 | 0.18114 |
| GHG (G) | 0.52649 | 0.47351 | 0.32813 |

Table 15 presents the ideal-solution distances $S$ and closeness coefficients $C$ for the four BWF World Tour schemes based on the TOPSIS method. The results show that Scheme D achieves the highest closeness value, driven by its strong performance in the high-weight indicators, particularly energy use and carbon emissions. Scheme A ranks second: although it has the lowest carbon emissions, it lags slightly in other dimensions. In contrast, Schemes B and C exhibit markedly lower closeness valuesespecially Scheme C and deviate substantially from the ideal solution, as illustrated in Figure 9.

## 4.4 Discussion: Infrastructure Strategies

Our entropy-TOPSIS results show that energy use and greenhouse gas emissions have the biggest influence on a citys final sustainability score. These two indicators vary the most across

Table 15: TOPSIS Results for BWF World Tour Schemes

| Scheme | $S^+$ | $S^-$ | Closeness $C$ | Rank |
|---|---|---|---|---|
| A | 0.296386 | 0.320137 | 0.519262 | 2 |
| B | 0.368197 | 0.208805 | 0.361879 | 3 |
| C | 0.423782 | 0.149165 | 0.260346 | 4 |
| D | 0.244496 | 0.306430 | 0.556209 | 1 |

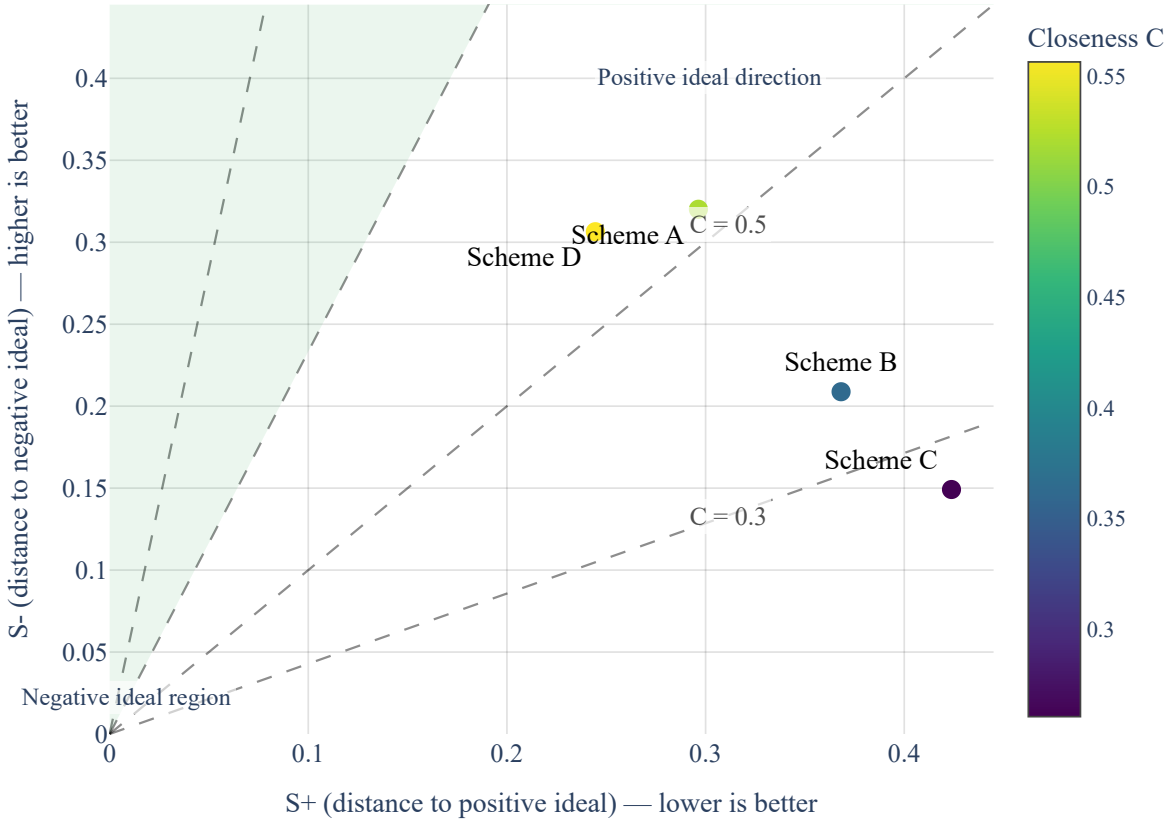



Figure 9: Visualization of TOPSIS Results for BWF World Tour Schemes

cities, so lowering $E$ and $G$ gives the quickest improvement for any future host.

In the short term, cities do not need major infrastructure changes to make progress. Stadiums can switch some lighting to LEDs, limit HVAC hours, and install basic water-saving devices. These steps are inexpensive but reduce electricity and water use during the event. Cities can also buy green electricity for the event period, which lowers carbon emissions without changing the entire power grid. For transportation, event shuttle buses, combined eventtransit tickets, clear walking routes, and stricter parking management can shift some spectators away from private cars. These measures are simple to operate and still cut emissions.

In the long term, the cities with the strongest competitiveness will be those whose basic infrastructure is already low-carbon. Dense, transit-oriented districts, convenient public transportation,

a higher share of renewable energy, and well-developed waste and water systems help improve all major indicators before any event is even planned. Because our evaluation method is data-based, these long-term investments help a city achieve high sustainability scores across many future bid-snot just for a single green event.

# 5 Conclusion

This study develops a sustainability assessment framework for evaluating the environmental impacts of major sporting events and applies it to Super Bowl host selection and other international competitions. The main conclusions are as follows:

1. First, we decompose event impacts into four dimensionsenergy use, greenhouse gas emissions, water consumption, and waste generationand construct quantitative sub-models for each. For Super Bowl LIX, the model results show: energy use of 426.7 MWh, water consumption of 15,323 t, GHG emissions of 825.8 t $CO_2$e, and waste generation of 75.2 t.

2. Second, using an entropy–TOPSIS evaluation model, we find that among six previous Super Bowl host cities (Glendale, Las Vegas, New Orleans, Santa Clara, Inglewood, and Atlanta), New Orleans exhibits the highest sustainability performance. Applying the same framework to four NFL home cities that have never hosted the event (Houston, Minneapolis, Indianapolis, and Charlotte), we find that Indianapolis ranks highest.

3. Finally, we extend the model from a single-city, single-game configuration to a multi-stage, multi-city format, using the BWF World Tour as an example. The extended framework accounts for resource use and emissions at each stage of the tournament, includes inter-city travel between host locations, and adds factors such as event frequency, venue reuse, and field-maintenance water use. Evaluation of four recent tour schemes shows that Scheme D achieves the best overall environmental performance.

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# 6 Letter

Dear Members of the National Football League,

We are writing to share the findings from our recent environmental assessment and to recommend the most sustainable host city for the Super Bowl in 2029. We developed an integrated model to assess energy use, water consumption, greenhouse gas emissions, and waste generation across multiple cities, identifying New Orleans, Louisiana, as the most sustainable host choice.

Our model shows that New Orleans performs well across all major environmental indicators. Among all candidate cities, it exhibits the lowest greenhouse gas emissions, supported by high energy efficiency, a compact and walkable urban layout, and a local grid with a high share of renewable energy.

In addition to its carbon advantages, New Orleans demonstrates strong performance in water and energy efficiency. The Caesars Superdome operates with lower-than-average energy and water demand compared with similar stadiums, and the city's existing public transit options help limit the environmental impact of spectator transportation. Because our evaluation places considerable weight on balanced environmental performance, New Orleans consistent strengths across all four indicators set it apart from other candidates.

Moreover, New Orleans requires little new construction to host the Super Bowl, which avoids the substantial environmental cost associated with building new facilities. Its mild February climate also reduces heating and cooling demands, further lowering the events energy footprint. These qualities make New Orleans not only a responsible choice, but also a practical and realistic one.

To conclude, we respectfully recommend New Orleans as the host city for the 2029 Super Bowl based on its strong sustainability profile and alignment with long-term environmental goals. Choosing New Orleans would allow the NFL to demonstrate leadership in promoting environmentally conscious event planning and set a meaningful example for other major sporting organizations.

Sincerely yours,

Your friends